Console Based Warehouse Management System

Team Name: WMSquad

Team Members:

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Project Description:

The Warehouse Management System (WMS) is a console-based Java application designed to help warehouse operators efficiently manage inventory, track orders, and oversee shipments. Through a simple text-based interface, users can monitor stock levels, process orders, and generate basic reports to streamline daily operations. The system aims to simplify warehouse management tasks and ensure that inventory and shipments are well-organized, improving overall efficiency in a straightforward manner.

Features/Functionality:

- **Inventory Tracking**: This feature monitors and manages stock levels. It allows for adding, updating, and removing items from the inventory, as well as tracking item quantities.
- **Order Processing:** This feature helps streamline the handling of orders. It validates the availability of requested items, reserves them for shipping, and processes the order for fulfillment.
- **Shipment Management:** This feature ensures that the movement of goods in and out of the warehouse is properly managed. It tracks shipments from the moment an order is fulfilled to the time it reaches its destination.
- **Generating Detailed Reports:** This functionality provides detailed analytics and insights into warehouse operations. The system generates reports related to inventory status, order fulfillment, shipments, as well as the total sales.

Scope and Limitation:

The scope of the Warehouse Management System (WMS) project focuses on core warehouse tasks such as tracking inventory, processing orders, managing shipments, and generating basic reports. It is designed to help warehouse operators perform daily tasks more efficiently through a simple, text-based interface. It can monitor stock levels, fulfill orders, and track shipments using straightforward commands. The system will also provide basic reports on inventory status and order history to assist with warehouse operations. However, the system's limitations include the absence of a graphical user interface (GUI), which may make it less intuitive for some users; a dedicated class for payments; a class for invoice. Due to limitations in programming knowledge, the WMS project also lacks error handling in some instances. Additionally, the system is designed for basic functionality and will not handle more complex warehouse tasks or integrate with larger business software.

Basic Classes:

- Scanner - This class essentially facilitates interaction between the system and the user, making it a crucial component for user-driven commands.
- Item - Represents an item with attributes like item ID, name, quantity, price, and location.
- Order - Represents customer orders with order ID, customer name, address, item ID, quantity, and total amount.
- Shipment - Handles the shipment of orders with tracking ID, order ID, and status.
- Report - Generates detailed reports for inventory, shipments, and orders, providing insights into stock levels, shipment status, and order fulfillment.
- WareHouseSquad - The main class that manages the warehouse operations and contains all the logic for managing items, orders, and shipments.

Class Diagram:

